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ORIGINAL RESEARCH

Drug approvals in crowded drug classes potentially reflect efficacy overestimates and false positivity: A portfolio analysis of immune checkpoint inhibitor trials

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Abstract

Objectives: Institutions making decisions regarding drug development, approval, or insurance reimbursement require accurate estimates of drug efficacy that reflect the drug's true value. Stein estimation and multiplicity correction can be applied to trials for an entire class of drugs, minimizing random error in effect size estimates.

Study Design and Setting: We downloaded ClinicalTrials.gov registry entries for trials testing immune checkpoint inhibitor (ICI) drugs, including only trials with overall survival (OS) or progression-free survival (PFS) as a primary outcome, in phases 1-3, with 100 patients or more, at least one US site and primary completion date before January 1, 2023. Effect sizes and *P* values were recalculated for pivotal trials. A change was considered ''meaningful'' if the adjusted hazard ratio (HR) crossed the null or an European Society for Medical Oncology (ESMO)-Magnitude of Clinical Benefit Score (MCBS) cutoff.

Results: We included 14 ICIs, eight of which were approved by the Food and Drug Administration (FDA). These were tested in 111 trials, 45 of which led to FDA approvals. Among FDA-approved drugs, 82 distinct OS/PFS HRs were reported as primary outcomes supporting an FDA label approval/revision. After recalculation, estimates improved to a meaningful degree for one pivotal trial (1%) but diminished meaningfully for 29 (35%). Among OS/PFS HRs, 74 (90%) had an original *P* value that was less than 0.05. Among these, 20 (27%) had a corrected *P* value that lost significance.

Conclusion: When statistically adjusting for the totality of trials testing ICIs, effect sizes in pivotal trials generally decreased, and some *P* values lost significance. This may have implications for minimizing error in approval, reimbursement, and certain clinical decisions.

Keywords: Bioethics; Research ethics; Clinical trial ethics; Clinical trial portfolios; Immune checkpoint inhibitors; Drug development

1. Introduction

Regulatory authorities like the United States Food and Drug Administration (FDA), clinical practice guideline committees and health technology agencies (HTAs) play key roles in public health. All three function as gatekeepers, regulating which drugs reach patients. In doing so, they also influence incentives for drug development. Accurate assessments of a drug's safety and efficacy are a key to bringing the supply of pharmaceutical innovation into alignment with their demand.

In deciding approval or reimbursement, drug regulators and HTAs typically rely on evidence produced in trials of a drug against a specific indication. Their task is to use this evidence to make the most accurate possible estimates of the drug's clinical value.

However, many drugs submitted to regulators and HTAs, or considered by clinical practice guideline committees, are

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Plain Language Summary

Drug approval and insurance reimbursement require accurate estimates of drug efficacy. In cases where drugs are tested in many different trials, statistical methods can be used to minimize error among efficacy estimates. In this paper, we use a statistical method to minimize error in calculations of the efficacy of drugs in a class of cancer drugs known as "immune checkpoint inhibitors" (ICIs). After this adjustment, there was a meaningful improvement in the estimate of efficacy for one (1%) of the clinical trials of ICIs that were used to justify FDA approvals, but estimates of efficacy for 29 (35%) of pivotal trials meaningfully worsened. We also corrected for potential false positives among trials in our sample. Among the 74 (90%) drug trials that had a significant result ($P < .05$), 20 (27%) lost significance after correction. The analyses performed herein have potential for reducing the potential bias in efficacy estimates used for making policy.

members of crowded drug classes ("crowdedness" is a matter of opinion; we suggest >5 drugs as a standard). When there are many different drugs in a class and they are tested against many conditions, random error produces some estimates that are spuriously large, and others that are spuriously small. If decisions are made using only evidence about a single drug, authorities may miss opportunities to absorb evidence from other similar drugs and minimize random variation. This may result in over-use of a drug, or it may amplify incentives to develop drugs in a crowded class rather than to pursue drugs in different classes. Therefore, there are good policy reasons to use evaluation methods that minimize error, not just for effect estimates of a single drug-indication pairing but for all trials within a drug class (a "drug portfolio") [1,2].

Immune checkpoint inhibitors (ICIs, see Table 1) for cancer represent a crowded class of drugs [3,4], with 10 approvals and at least 33 in development [5]. Many are directed toward the same target [6]. A 2022 analysis reported that over 2000 trials testing ICIs were open [5]. A search of ClinicalTrials.gov for trials testing approved immunotherapy drugs returned 5949 trials, with a combined enrollment of 1,599,560 patients. Such intense commitment to a narrow set of clinical hypotheses raises the prospect that extreme effects in individual trials used for regulatory approvals or HTA may partly reflect random variation. Such intense work within a portfolio may also crowd out more innovative and impactful drug development efforts outside the drug class.

In what follows, we apply a statistical technique to minimize error in the portfolio of ICI cancer trials and study the potential effect on approvals or treatment recommendations. Stein estimation was first developed in 1955 [7] and is widely used to minimize estimation error in large datasets where multiple quantities are estimated. Stein estimation can adjust outcomes in one estimate (eg, a phase 3 trial leading to drug approval) based on outcomes of other trials in the same portfolio. Stein estimation is routinely used in diverse areas of medical research to reduce error in estimating vaccination coverage [8], dose-response statistical modeling [9], and others [7,10].

We explore how efficacy for pivotal trials would change, were such state-of-the-art statistical techniques applied to the portfolio of ICI drug trials. We also used European Society for Medical Oncology (ESMO)-Magnitude of Clinical Benefit Scores (MCBSs) to determine how assessments of clinical value would change.

2. Methods

2.1. Data collection of pivotal evidence

To capture all ICIs tested in cancer, we performed a scoping review of review articles [11]. We then obtained drug name synonyms from the US National Library of

Table 1. Abbreviations used in this document

Abbreviation	Full
BTC	Biliary tract cancer
CC (MSI-H)	Colorectal cancer (microsatellite instability-high)
CHL	Classical Hodgkin lymphoma
EC	Esophageal cancer
ESCC	Esophageal squamous cell carcinoma
ESMO-MCBS	European Society for Medical Oncology Magnitude of Clinical Benefit Scale
FDA	Food and Drug Administration
GC	Gastric cancer, Gastroesophageal Junction Cancer And Esophageal Adenocarcinoma
HC	Hepatocellular carcinoma
HNSCC	Head and neck squamous cell carcinoma
ICI	Immune checkpoint inhibitor
MPM	Malignant pleural mesothelioma
NIH	National Institutes of Health
NSCLC	Nonsmall cell lung cancer
NSNSCLC	Nonsmall nonsquamous cell lung cancer
NSSCLC	Nonsmall squamous cell lung cancer
OS	Overall survival
PFS	Progression-free survival
RCC	Renal cell carcinoma
SCLC	Small cell lung cancer
TNBC	Triple-negative breast cancer
UC	Urothelial carcinoma

What is new?

Key findings

- Portfolio-level Stein shrinkage reduced effect size estimates for 35% of pivotal OS/PFS hazard ratios supporting FDA approvals, while only 1% improved.
- After controlling for family-wise error, 27% of pivotal trials with $p < 0.05$ lost statistical significance.

What this adds to what is known?

- This study demonstrates, using an entire crowded drug class (immune checkpoint inhibitors), that borrowing strength across trials can systematically reduce overestimation in regulatory-grade evidence.

What is the implication and what should change now?

- Regulators and reimbursement authorities should consider routine portfolio-level shrinkage and multiplicity correction when evaluating drugs in crowded classes.
- Incorporating these methods could reduce false positives, temper inflated benefit estimates, and better align pricing and innovation incentives with true clinical value.

Medicine's ChemIDplus (https://chem.nlm.nih.gov/chemidplus/) (see Table 1 in the study protocol) [12]. For each FDA-approved use of an ICI in cancer, we extracted evidence cited in the publicly available drug labels section14 (https://www.accessdata.fda.gov/scripts/cder/daf/) using a structured data collection form, with data cutoff on January 1, 2023.

We extracted the pivotal evidence reported at the time when the approved use was first mentioned on a drug label. For each approved use, we determined the major outcomes, supporting the approval and extracted hazard ratios (HRs) of overall survival (OS) and progression-free survival (PFS).

2.2. Sampling of ICI trials

We aimed to obtain the full sample of eligible trials belonging to the ICI trial portfolio from ClinicalTrials.gov. We searched ClinicalTrials.gov for registered trials that tested an ICI in cancer, registered OS/PFS as a primary outcome in phases 1, 2, or 3, with an enrollment of 100 patients or greater (to exclude trials that are unlikely to have influenced policy decisions, such as pilot or early-phase studies), recruited in the United States, and listed actual primary completion before January 1, 2023, and a trial status as "active, not recruiting," "completed," "suspended," or "terminated" (see Appendix 2 for details) [12].

We did not include trials if they did not report any relevant test statistics or if test statistics were uninterpretable. For OS/PFS, we excluded trials without HRs and confidence intervals (CIs) or *P* values and cohort size per arm (eg, trials reporting only median OS/PFS).

2.3. Outcomes for ICI trials

For eligible trials, efficacy data were manually extracted from ClinicalTrials.gov. In cases where data were not reported on ClinicalTrials.gov, we extracted data from journal articles linked on the ClinicalTrials.gov record, if any. These data were double-extracted by two independent coders with disagreement resolved by discussion. When multiple HR estimates were reported for the same end point, eg, in the case of multiple OS HRs to report comparisons between arms of studies with more than two arms, all HRs were collected. Subgroup analyses were not included.

The present study was not aimed at estimating the effect of one intervention by pooling results of all trials and testing the effect. Instead, the goal was to leverage portfolio analysis to examine how shrinkage affected individual estimates. Therefore, we deemed risk of bias analyses normally presented in synthetic studies like ours unsuited to our objectives.

ESMO score cards were obtained for all pivotal trials in our sample by searching and matching them by drug, indication, and the identifier for the relevant trial (https://www.esmo.org/guidelines/esmo-mcbs/esmo-mcbs-for-solid-tumours).

2.4. Portfolio analysis

Our primary outcomes were i) to estimate the magnitude of effect size shrinkage among the trials reporting OS/PFS HRs that were used to justify new indications on FDA labels for ICI drugs and ii) to estimate the rate of false positivity among these trials, based on a statistical correction for their inclusion in a larger portfolio of trials. As a secondary outcome, we also revised ESMO-MCBS based on our shrinkage-corrected estimates of effect sizes.

For the primary outcomes, we operationalized i) the impact of shrinkage based on whether point estimates for effect sizes changed in a manner that was potentially significant, clinically and ii) whether *P* values that were originally reported as less than 0.05 exceeded 0.05 after correction. The secondary outcome was based on a reanalysis of ESMO scores. For pivotal trials in our sample with available scorecards for the MCBS available from ESMO [13], the score was revised to take the shrinkage-corrected effect sizes into account by an oncologist (JDP).

ESMO-MCBS are based on several factors, one of which is the HR for the estimate of treatment effect. For OS, there are three HR "cutoffs" at which an MCBS might change—0.65, 0.70, and 0.75—limits used in ESMO-MCBS evaluation forms to distinguish in certain cases between grades. Similarly for PFS, there is one HR ‘‘cutoff’’ at which an MCBS might change, 0.65. We will refer to HRs that cross parity or one of the MCBS grading cutoffs when shrinkage is applied to have been ‘‘meaningfully’’ improved or degraded, depending on the direction of the change.

2.5. Data and code

Data and code are available on Open Science Framework (OSF) [12]. Effect size shrinkage was performed using the bayesmeta package in *R*.

2.6. Statistics

Shrinkage estimation was performed using the bayesmeta function from the eponymous package in R to perform a random-effects meta-analysis of the entire portfolio of ICI trials. This reanalysis depends on the observation that within a portfolio of clinical trials, there are two sources of variability—both the underlying heterogeneity of treatment effects due to differences in the trials and random variation in each of the observed effects. Thus, the results selected from a portfolio of trials based on a larger treatment effect, such as pivotal trials, likely represent an overestimate of the real effect. This can be corrected by ‘‘shrinking’’ the observed effects toward the mean treatment of the portfolio. Optimal shrinkage, calculated here with Stein estimation, is a function of the magnitudes of the two sources of variation [1]. This method can be used to combine studies to estimate the overall mean effect size and between-study heterogeneity based on the individual trials’ HRs and standard errors. We are not interested in the summary effect size estimate, but rather we used this method to derive a shrinkage-corrected estimate for each study’s OS/PFS HR, which is informed by ‘‘borrowing strength’’ from the remaining studies [14]. We used a uniform prior for the overall mean effect and a half-normal prior for the heterogeneity. As a sensitivity analysis, we reanalyzed shrinkage, changing the sigma two-fold and four-fold, and found no change to the interpretation of our results (see Appendix section D [12] for details).

For false positivity correction, we applied a parametric bootstrap procedure to control the probability of family-wise error rate (FWER), ie, at least one false rejection, using the extracted OS/PFS HRs and the corresponding 95% CI from the pivotal trials in the ICI drug trial portfolio [15]. We hypothesized that the null distribution mean of the log HR is zero, and for each individual trial, we calculated the log of standard error of the HR. To define a reference distribution conforming to the global null hypothesis of no effect, we used a log-normal distribution with the mean given by *log(HR)* and shifted the distribution to be centered around the assumed null effect. We sampled, with replacement, 10,000 random values from the reference distribution and counted the number of values as extreme or more extreme as the observed *log(HR)* and divided this sum by the number of resampled values (10,000) to get the FWER-adjusted *P* value. FWER adjustment is different from shrinkage and is concerned with protecting the Type I error due to multiple statistical tests conducted (one per study in the portfolio). By adjusting the *P* values for FWER, we ensure that even when there is no treatment effect in any of the studies in the portfolio, the probability of incorrectly finding at least one significance is at most 5%. This adjustment depends on the unadjusted study *P* values and the size of the portfolio.

2.7. Protocol deviations

Our protocol was prespecified and registered with OSF [12]. Data collection cutoffs were extended from January 1, 2022, to January 1, 2023, to update the dataset after unforeseeable work disruptions. We had originally planned to implement a second level of protection against inflated Type I errors in portfolio analysis using the null interval method [16]. This analysis was later deemed inappropriate and dropped before implementation.

3. Results

3.1. Portfolio of ICI cancer trials

We included 14 ICI drugs, eight of which were approved by the FDA and six of which were not FDA-approved by the time of data cutoff (see Table 2). These were tested in 111 trials, 45 of which led to FDA approvals, and 66 of which did not at the time of data cutoff (see Table 3 for sample characteristics). Among FDA-approved drugs, 82 distinct OS/PFS HRs were reported as major or primary outcomes, supporting an FDA label approval or revision.

Among ICI trials not supporting an approval, 148 OS/PFS HRs were reported. Figure 1 contains a flowchart for trial inclusion, and Appendix Table 1 [12] contains a listing of all drug-indication pairings for FDA approvals and the number of pivotal trial results (OS/PFS HRs) used to justify them.

3.2. Effect size shrinkage

In Figures 2 and 3, we present the OS and PFS HRs among pivotal trials for drugs in our sample, which are arranged in order of the original HR. After applying shrinkage to 47 OS and 35 PFS HRs, point estimates for effect sizes improved to a meaningful degree for one PFS HR (1% of 82 OS/PFS HRs) and were degraded meaningfully for 29 pivotal trials (35% of 82 OS/PFS HRs). In particular, 26 OS HRs (32%) and three PFS HRs (9%) were degraded meaningfully.

Among the 26 OS HRs that decreased meaningfully, four (15% of 26) decreased to greater than 0.65, 14 (54%) decreased to greater than 0.70, and eight (31%) decreased to greater than 0.75.

No effect sizes shrank to the point where CIs included a null effect. Exact values for the original and shrinkage-

Table 2. Immune checkpoint inhibitor drugs in our sample

Drug	Immune checkpoint	FDA approved[a]	First FDA approval date
Ipilimumab	CTLA-4	Yes	March 25, 2011
Pembrolizumab	PD-1	Yes	September 4, 2014
Nivolumab	PD-1	Yes	March 4, 2015
Atezolizumab	PD-L1	Yes	May 18, 2016
Avelumab	PD-L1	Yes	March 23, 2017
Durvalumab	PD-L1	Yes	May 1, 2017
Cemiplimab	PD-1	Yes	September 28, 2018
Tremelimumab	CTLA-4	Yes	November 10, 2022
BMS-936559	PD-L1	No	-
Camrelizumab	PD-1	No	-
Pidilizumab	DLL-1	No	-
Sintilimab	PD-1	No	-
Tislelizumab	PD-1	No	-
Toripalimab	PD-1	No	-

[a] FDA approval as of data cutoff, January 1, 2023.

corrected HRs for all trials in our sample are reported in Appendix Tables 2 and 3 and presented in Appendix Figures 1 and 2 [12].

3.3. FWER correction

We recalculated P values for all pivotal trials in our sample to control for FWER (see Appendix Table 4 [12]).

Table 3. Sample characteristics

	Pivotal trials	Nonpivotal trials
Phase		
Phase 1	0 (0.0%)	1 (1.5%)
Phase 2	1 (2.2%)	13 (19.7%)
Phase 3	44 (97.8%)	52 (78.8%)
Funded by		
Industry	45 (100%)	65 (98.5%)
Other/NIH	0 (0.0%)	1 (1.5%)
Median enrollment (range)	790 (272–2748)	602.5 (100–1351)
Endpoints reported		
OS	37 (82.2%)	61 (92.4%)
PFS	27 (60.0%)	58 (87.9%)
ICI drug		
Atezolizumab	8 (18%)	13 (20%)
Avelumab	1 (2%)	8 (12%)
Cemiplimab	2 (4%)	1 (2%)
Durvalumab	5 (11%)	4 (6%)
Durvalumab + Tremelimumab	0 (0%)	1 (2%)
Durvalumab ± Tremelimumab	0 (0%)	1 (2%)
Ipilimumab	4 (9%)	9 (14%)
Nivolumab	10 (22%)	11 (17%)
Nivolumab or Ipilimumab	0 (0%)	3 (5%)
Pembrolizumab	15 (33%)	9 (14%)
Pembrolizumab or Ipilimumab	0 (0%)	2 (3%)
Tislelizumab	0 (0%)	1 (2%)
Tremelimumab	0 (0%)	3 (5%)
Total	45	66

NIH, National Institutes of Health; OS, overall survival; PFS, progression-free survival; ICI, immune checkpoint inhibitor.

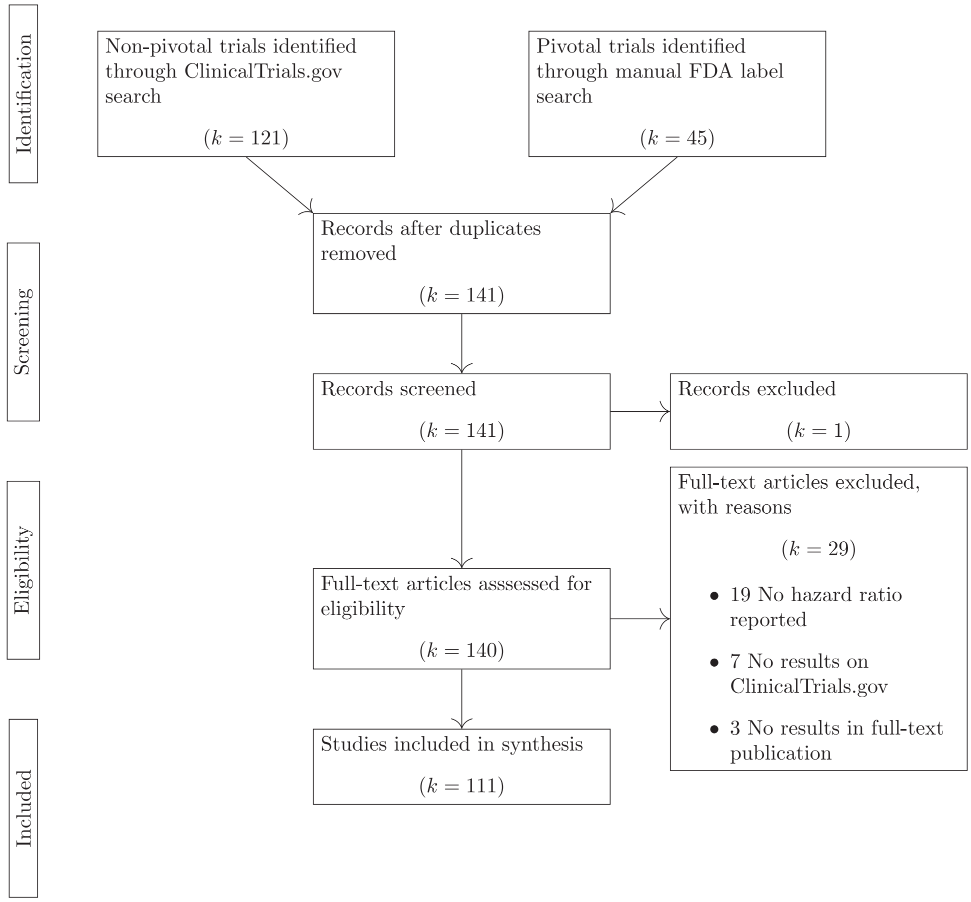

Figure 1. Flowchart for trial inclusion.

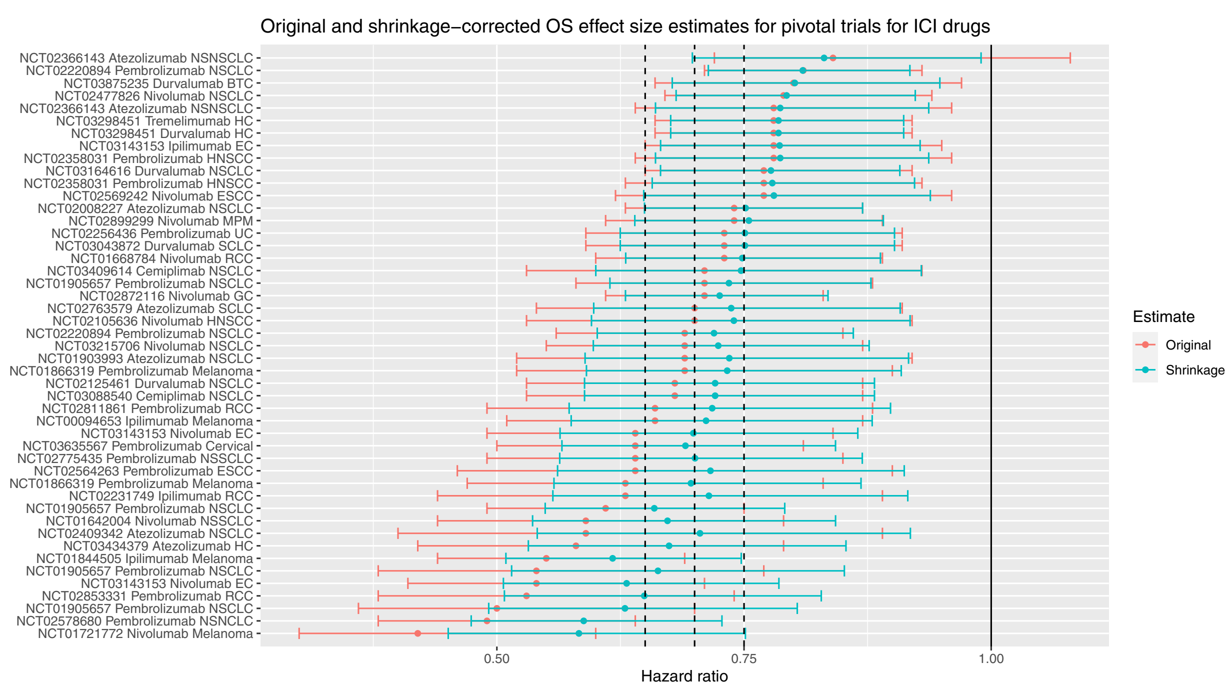

Figure 2. Original (red) and shrinkage-corrected (cyan) overall survival hazard ratios for pivotal trials for ICI drugs. Solid vertical line indicates no advantage to experimental treatment; dashed vertical lines indicate cutoffs used for ESMO-MCBS (0.65, 0.70, and 0.75). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article). MCBS, Magnitude of Clinical Benefit Scores; ESMO, European Society for Medical Oncology; ICI, immune checkpoint inhibitor.

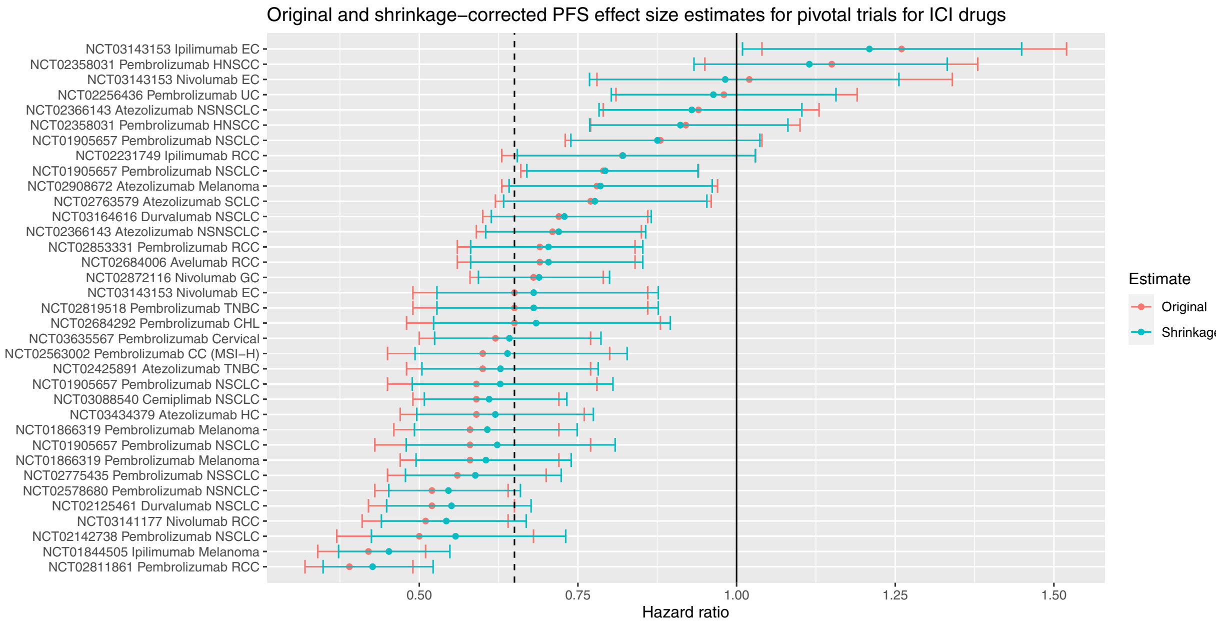

Figure 3. Original (red) and shrinkage-corrected (cyan) progression-free survival hazard ratios for pivotal trials for ICI drugs. Solid vertical line indicates no advantage to experimental treatment; dashed vertical line indicates 0.65 cutoff for ESMO-MCBS. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article). MCBS, Magnitude of Clinical Benefit Scores; ESMO, European Society for Medical Oncology; ICI, immune checkpoint inhibitor.

Among OS/PFS HRs that were used to justify FDA approvals in ICI drugs, 74 (90%) had an originally reported a *P* value that was less than 0.05. Among these, 20 (27%) had a FWER-corrected *P* value that exceeded 0.05.

Similarly, we also report recalculated *P* values for nonpivotal trials in our sample (see Appendix Table 5 [12]). Among OS/PFS HRs in trials of ICIs that were not used to justify FDA approvals, 45 (30%) had an originally reported *P* value that was less than 0.05. Among these, 22 (49%) had a FWER-corrected *P* value that exceeded 0.05.

Some tumors are more immunogenic than others. As noted by a referee, some policy-makers might be more concerned with minimizing error for tumor subsets that are more immunogenic. Therefore, as a post hoc sensitivity analysis, we recalculated OS and PFS HR shrinkage stratifying by indication, such that strength is only borrowed from trials within the same indication. Under this reanalysis, one MCBS would not have been revised downward (ipilimumab in EC). This is presented in Appendix Section E.

3.4. ESMO clinical benefit scoring

Among 82 pivotal OS/PFS HRs, 47 had corresponding published ESMO scorecards, 38 based on OS and nine based on PFS. Nine were given an ESMO-MCBS of 3 (noncurative, on this scale, 4 to 5 is considered ‘‘substantial benefit’’), 27 were scored as 4, and 11 were given a Clinical Benefit Score of 5. Appendix Tables 2 and 3 contain the original and revised MCBS for drug-indication pairings that were scored by ESMO [12].

All MCBS for the 35 FDA drug-indication pairs that were based on PFS and scored by ESMO remained the same (see Appendix Table 3 [12]). Among MCBS based on OS, two interventions (8%) were downgraded for clinical impact. In particular, pembrolizumab in head and neck squamous cell cancer (HNSCC) no longer qualified as offering substantial benefit (changed from 4 to 3) as was also the case for ipilimumab in esophageal cancer (EC) (changed from 4 to 2; see Appendix Table 2 [12]).

4. Discussion

Adjusting for the totality of trials and testing ICIs generally led to decreased effect size estimates for pivotal trials supporting approval. Effect estimates for about a quarter of pivotal trials lost statistical significance after controlling for FWERs. In two cases (ipilimumab for EC and pembrolizumab for HNSCC), the ESMO-MCBS would be revised downward if shrinkage-corrected effects were used to grade the ICI drug-indication pairings.

Stein estimation offers an approach for reducing errors in a drug class characterized by intense trialing. It may seem counterintuitive that an efficacy estimate for a trial testing atezolizumab against nonsmall cell lung cancer (NSCLC) could be meaningfully combined with a trial-testing nivolumab against renal cell carcinoma (RCC). However, Stein estimation provides a sound method of minimizing error, and it can be exploited to prevent overestimation in treatment benefit [1,7,14]. Readers who are puzzled by the seeming paradoxical nature of the approach can find an accessible review here [7].

While we have mostly discussed shrinkage that diminishes OS/PFS estimates of an ostensibly effective drug, a trial with an effect estimate that seemed unpromising may be shrunk toward the portfolio mean and away from a null, suggesting that an apparently negative trial result may warrant a closer look.

An FDA approval can set off a wave of research into a drug class or mechanism [17−19]. Bodies like FDA have a mandate to promote public health through innovation [20,21]. Therefore, they have a stake in accurately signaling the efficacy of drugs as members of their drug class as they are approved. Shrinkage correction of effect sizes allows for minimization of error in a class. When sponsors run a large numbers of trials testing a new drug against different indications, random variation alone may produce large effect estimates. Routine use of portfolio analysis by regulatory authorities, like that performed above, might check efforts by sponsors to exploit random variation by running a large numbers of trials of a drug again in many indications.

Reimbursement authorities also have interests in controlling resources flowing to a single class of drugs, especially when drugs in other classes present similar clinical value. Reimbursement authorities and formularies can use information about outcomes in trials of the same class to reduce overall error and make better recommendations about a drug. Our study shows that doing so may lead to adjustments in terms of what is offered. To be clear, however, the effects of shrinkage are buffered by the larger sample sizes in pivotal trials, and the adjustments we performed do not lead to radical changes in estimates. The methods we use may have the greatest impact on cost estimates for new drugs, which are based on efficacy estimates that would mostly diminish under a shrinkage analysis.

This study has several limitations. First, our portfolio contained trials of ICI drugs in the USA with OS/PFS primary end point and greater than 100 patients. Different inclusion criteria for a portfolio might produce different results. We might have narrowed our analysis to look only at trials of different drugs and drug combinations for a single indication. Indeed, we invite readers to use our code and estimates to perform their own analyses. Second, our analysis drew on pivotal trials from the FDA only and shrank estimates only from trials registered on ClinicalTrials.gov. The analysis is thus US-centered. Third, our analysis was based on estimates posted to ClinicalTrials.gov. Some estimates that we used to conduct shrinkage analysis would not have been available to regulators or HTAs when they were making approval or reimbursement decisions. As well, we did not assess secondary publications for more mature survival evidence. Fourth, the present study—like many in meta-analysis—is novel and does not correspond to a reporting guideline type. None were followed.

Finally, while shrinkage can reduce error, it cannot infallibly reveal false positives or predict regulatory reversals. Conclusions about clinical interpretability of shrunken trial effect sizes should be made with care. Our analysis applied shrinkage with a ‘‘policy lens,’’ie, to a wide set of estimates policymakers use. A clinician would typically be focused on a much smaller subset of estimates when deciding which treatment to use. There, one would expect less dramatic shrinkage. Among the FDA labels for drugs in our sample, there was only one indication approved based on OS/PFS that later had FDA approval withdrawn by data cutoff: atezolizumab for triple-negative breast cancer (TNBC) (https://www.fda.gov/drugs/resources-information-approved-drugs/withdrawn-cancer-accelerated-approvals; other indications approved for atezolizumab and later withdrawn were based on outcomes other than PFS or OS, such as response rates). Our analysis produced a shrinkage of 0.60 to 0.63 for PFS in a trial of atezolizumab in TNBC. This decrease was neither so large that it crossed an MCBS grading threshold nor did our study flag this trial as a potential false positive based on a FWER correction.

5. Conclusion

The approval of drugs by the FDA depends on accurate assessments of efficacy reported in trials. Effect sizes for drugs can be more accurately assessed by statistically correcting for portfolio-level misestimation in both effect size and false positivity. This has implications for regulatory decision-making, interpretation of trials, and reimbursement policy.

CRediT authorship contribution statement

Benjamin Gregory Carlisle: Writing − review & editing, Writing − original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Mithat Gönen:** Methodology, Formal analysis, Conceptualization. **Luciano Fernandez:** Writing − review & editing, Data curation. **Joseph C. Del Paggio:** Writing − review & editing, Investigation, Formal analysis, Data curation. **Jonathan Kimmelman:** Writing − review & editing, Writing − original draft, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

Jonathan Kimmelman received consulting fees from Novartis.

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Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jclinepi.2026.112182.

Data availability

Data are on OSF, link in manuscript.

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